A quartic-width learning bound for Bayesian networks stable under internal noise

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Abstract

We give an explicit sample-complexity bound for Gaussian Bayesian tanh networks restricted by stability of clipped predictions under simultaneous internal disturbances. For a layered Boolean target circuit of width C, the sufficient number of examples is $O(C^4 \log^9(2+C))$ when the other parameters are fixed. The dependence on depth is polynomial, and the statistical accuracy factor is $1/\varepsilon$. The guarantee holds at every learner width above the teacher width in the strong-prior regime, and at every actual finite-width Gaussian scaling with exponent $\gamma \geq 2$. The proof adapts an earlier pruning-and-Bernstein compression argument to the clipped-score prior and supplies explicit Gaussian posterior thresholds. It does not establish quadratic dependence on teacher width, or this improved rate at mean-field scaling. The proof has independent AI-agent mathematical review but is not Lean verified.

1 Model and purpose

A polynomial-depth learning theorem is more useful when the polynomial has a manageable degree. This paper extracts an explicit quartic dependence on Boolean teacher width for a particular Gaussian neural-network prior, without strengthening its existing internal-noise support condition. The proof uses the deterministic compressor in the earlier depth report [1]; that compressor is not new here. The contributions are its adaptation to clipped-score stability, explicit rate bookkeeping, and the finite-scale posterior conclusion.

Let $d \geq 1$, $L \geq 2$, $H = L - 1$, and $\mathcal{X} = \{-1, 1\}^d$. The network has H tanh hidden layers, all of width N, and a linear readout:

$$h^1 = \tanh(W^1 \bar{x}), \qquad h^s = \tanh(W^s h^{s-1}), \qquad z_w = v^\top h^H, \qquad \bar{x} = (x, 1).$$

The independent Gaussian entries have variances t^2/d in W^1, t^2/N in internal matrices, and t^2/N^2 in v, for $0 < t \leq 1$. Write

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{H} \|W^s\|_F^2 + N\|v\|_2^2, \qquad E_v = N\|v\|_2^2.$$

The scale $t = N^{(1-\gamma)/2}$ corresponds to exponent γ.

Disturbances are added after every hidden activation and at the output:

$$h_\eta^1 = \tanh(W^1 \bar{x}) + \eta^1, \quad h_\eta^s = \tanh(W^s h_\eta^{s-1}) + \eta^s, \quad z_w(x; \eta) = v^\top h_\eta^H + \eta^{\text{out}},$$

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with $\|\eta\|_*^2 = \sum_s \|\eta^s\|_2^2/N + |\eta^{\text{out}}|^2$. Let $T(u) = \max\{-1/2, \min\{1/2, u\}\}$ and fix $c > 0$. The prior conditions the raw Gaussian $\mu_{N,t}$ on the event

$$\mathcal{R}_{N,c} = \left\{ w : |T(z_w(x;\eta)) - T(z_w(x))| \le 1/16 \quad \text{for every } x,\ \|\eta\|_* \le r_c(w) := \frac{c}{\sqrt{1 + HE_v(w)}} \right\}.$$

The noisy hidden values need not lie in $[-1, 1]$. For iid inputs $X_i \sim D$, training further conditions on $f(X_i)z_w(X_i) > 1/2$. Denote this posterior by $Q_{S,N,t,c}$. The risk is $R(Q) = \mathbb{E}_{w\sim Q, X\sim D}\mathbf{1}\{f(X)z_w(X) \le 1/4\}$, which bounds ordinary sign error. No initial clean-output margin is required at unobserved inputs.

The target f is computed by a layered Boolean circuit of width C and H stages, using NAND, copy, and negation gates. Constant wires and any compilation overhead count toward width; its output is one final wire. Idle stages can be padded with copies. This is a Boolean-circuit comparator, not an arbitrary real-weight neural implementation.

2 An explicit learning theorem

The target construction in Appendix A gives an explicit $D_* = D_*(d, L, C, c) \ge 1$ such that, for every $N \ge C$, a subset of $\mathcal{R}_{N,c}$ has correct margin on the entire cube and raw Gaussian mass at least e^{-ND_*/t^2}. In particular, all posteriors used below are well-defined.

For $K \ge 1$, define

$$r = \min\{1/6, c/\sqrt{1 + HK}\}, \qquad m = \left\lceil 256Kr^{-2}[d + \log(400L)] \right\rceil,$$
$$B = \sqrt{4mK}, \qquad u = \frac{1}{48(d + 1 + Lm)(1 + m(B + 1))^L},$$
$$P = m(d + 1) + (L - 2)m^2 + m, \qquad J = P \log(3 + 2B/u).$$

Theorem 2.1. *For $0 < \varepsilon, \delta < 1$, it suffices to take*

$$n \ge \frac{2}{\varepsilon}[J + \log(1/\delta)]. \tag{1}$$

The following choices of K and scales are valid:

1. *Set $K = 8D_*$. At every $N \ge C$, allow all scales satisfying*

$$t^2 \le \min\left\{1, \frac{ND_*}{(P_N/2)\log 2 + \log(2/\varepsilon)}\right\}, \quad P_N = N(d + 1) + (L - 2)N^2 + N.$$

2. *Set $K = 8D_* + 4(d + L) + 4\log(2/\varepsilon)$. Allow every $N \ge C$ and $0 < t \le N^{-1/2}$, including every exponent $\gamma \ge 2$.*

With probability at least $1 - \delta$ over the sample, $R(Q_{S,N,t,c}) \le \varepsilon$ simultaneously over the indicated widths and scales.

For the original strong-prior parameter σ, one has $t = \sqrt{N}\sigma$. The first clause supplies a sufficiently small positive threshold at every fixed width. The second clause concerns actual finite-variance priors. The sample event is independent of width and scale in both clauses.

The code budget satisfies

$$J = O\Big[\Big(dLKr^{-2}(d + \log L) + L^2K^2r^{-4}(d + \log L)^2\Big)\log(2 + d + L + K + r^{-1})\Big].$$

The appendix's formulas give, with a universal constant,

$$D_* \le O\Big((d+LC)[\log^2(2C)+\log(2L)+\log(2+d+C)]+(1+c^2)C\Big).$$

Consequently the sample bound is polynomial in depth and $O(C^4\log^9(2+C))$ when the remaining parameters are fixed. The exact sample formula retains the $1/\varepsilon$ factor and the additional logarithmic dependence entering the finite-scale energy cutoff.

3 Uniform compression on an energy sublevel

Fix w with $\kappa_N(w) \le K$. Put $k_s = \|W^s\|_F^2$ for internal layers and $k_L = N\|v\|_2^2$, so their sum is at most K. In every internal row, and in the readout, delete entries with magnitude larger than

$$\frac{8\sqrt{K}\,\|\mathrm{row}\|_2}{r\sqrt{N}}.$$

Zero rows remain zero. Denote the pruned weights by bars. The total absolute mass removed from a nonzero row is at most its squared norm divided by the threshold. Therefore, on any vector in $[-1,1]^N$, the internal pruning error has norm at most $r\sqrt{Nk_s/K}/8$; the readout error is at most $r\sqrt{k_L/K}/8$.

Independently choose m uniform indices S_j^s with replacement at every hidden layer. Keep the selected first-layer rows and multiply every subsequent sampled source weight by N/m. The virtual full-width pass is

$$\begin{aligned}
\widetilde{h}^1 &= \tanh(W^1\bar{x}),\\
\widetilde{h}_i^s &= \tanh\left(\frac{N}{m}\sum_{j=1}^{m}\bar{W}^s_{i,S_j^{s-1}}\widetilde{h}^{s-1}_{S_j^{s-1}}\right),\\
\widetilde{z} &= \frac{N}{m}\sum_{j=1}^{m}\bar{v}_{S_j^H}\widetilde{h}^H_{S_j^H}.
\end{aligned}$$

Its stored coordinates form an ordinary width-m network, including when indices repeat or $m > N$.

Conditional on $S^1,\dots,S^{s-2}$, the virtual state $\widetilde{h}^{s-1}$ is independent of the fresh list S^{s-1}. For one uniform source index I, put $Y = N\bar{W}^s_{:,I}\widetilde{h}^{s-1}_I$. The variance and norm bounds are

$$\mathbb{E}\|Y-\mathbb{E}Y\|_2^2 \le Nk_s, \qquad \|Y-\mathbb{E}Y\|_2 \le 16\sqrt{NKk_s}/r.$$

Hilbert-space Bernstein [3], with tolerance $r\sqrt{Nk_s/K}/8$, bounds failure by

$$2\exp\left(-\frac{3mr^2}{640K}\right) \le 2\exp\left(-\frac{mr^2}{256K}\right).$$

The readout calculation is identical without factors $\sqrt{N}$. Zero-energy stages have zero error. A union bound over the 2^d inputs and at most L stages makes total failure at most $1/200$ by the chosen m.

On success, pruning and sampling together have normalized stage error at most $r\sqrt{k_s/K}/4$. Define disturbances in the original network by

$$\eta^1 = 0, \qquad \eta^s = \widetilde{h}^s - \tanh(W^s\widetilde{h}^{s-1}), \qquad \eta^{\mathrm{out}} = \widetilde{z} - v^\top\widetilde{h}^H.$$

Since tanh is 1-Lipschitz, the original disturbed network produces the virtual states exactly and, simultaneously at every input,

$$\|\eta\|_*^2 \le \sum_s \frac{r^2 k_s}{16K} \le r^2/16.$$

No product of downstream operator norms is required in this comparison.

The expected energy of the sampled network is at most K:

$$\frac{d}{m}\mathbb{E}\|W^1_{\mathrm{comp}}\|_F^2 = \frac{d}{N}\|W^1\|_F^2, \qquad \mathbb{E}\|W^s_{\mathrm{comp}}\|_F^2 \le \|W^s\|_F^2, \qquad m\mathbb{E}\|v_{\mathrm{comp}}\|_2^2 \le N\|v\|_2^2.$$

These identities use independent index lists in adjacent layers; pruning can only decrease the corresponding expectations. Markov's inequality gives compressed energy at most $4K$ with probability at least $3/4$. Its intersection with the all-input noise event is nonempty.

If $w \in \mathcal{R}_{N,c}$, then $r \le r_c(w)$. Choose a sampled network in that intersection. Its clipped output differs from the original's by at most $1/16$ everywhere, and every weight has magnitude at most $B = \sqrt{4mK}$. Round its weights to a fixed grid with coordinate error at most u. A forward Lipschitz recurrence, using rounded weights at most $B+1$, bounds the output change by

$$u(d+1+Lm)(1+m(B+1))^L \le 1/48.$$

Clipping is 1-Lipschitz. Thus the rounded network has clipped discrepancy at most $1/12$ everywhere. There are at most e^J rounded networks. This finite class is independent of N, t, and of the original network.

4 Generalization and the posterior energy tail

Use the ramp ψ which is one at signed clipped scores at most $1/3$, zero at scores at least $5/12$, and linear in between. If a robust network of energy at most K fits every training margin, its associated rounded network h has $\psi(f(X_i)T(h(X_i))) = 0$ on all training points. Every original test-margin failure has compressed ramp loss one. For any fixed h with population ramp loss greater than $\varepsilon/2$, the probability of zero empirical ramp loss is at most $e^{-n\varepsilon/2}$. Indeed, the probability of positive loss is at least its mean. A union bound over the finite class and (1) show that every robust fitting network of energy at most K has risk at most $\varepsilon/2$, on one event of sample probability at least $1-\delta$. This event is simultaneous over widths and applies to all posterior mixtures supported on that energy sublevel.

Under the raw Gaussian, $N\kappa_N/t^2$ has the chi-square distribution with P_N degrees of freedom. Its exponential moment and the target cylinder's mass give, uniformly over training samples,

$$Q_{S,N,t,c}\{\kappa_N > K\} \le \exp\left[-\frac{NK}{4t^2} + \frac{P_N}{2}\log 2 + \frac{ND_*}{t^2}\right]. \tag{2}$$

For $K = 8D_*$, the first scale threshold in Theorem 2.1 makes this at most $\varepsilon/2$. If $t^2 \le 1/N$, then $t^2 P_N/N \le d+L$. Multiplying the exponent by t^2/N bounds it by

$$-K/4 + D_* + (d+L)(\log 2)/2 \le -\log(2/\varepsilon)$$

for the second choice of K. Since $N/t^2 \ge 1$, the tail is again at most $\varepsilon/2$. The retained-risk bound plus this discarded mass proves the theorem.

5 Scope and provenance

The quartic rate was already implicit in the energy-compression argument of the 17 October depth report [1]. The present argument adapts that proof to the clipped-score support and makes its finite Gaussian consequence explicit. The target construction below is reproduced from the robust-prior paper [2], including work to which Claude contributed. The review records accompanying this paper identify the independent AI-agent mathematical checks; this paper is not Lean verified and has not received a full human mathematical review.

At mean-field scaling, the expected internal Frobenius energy grows with width; for $1 < \gamma < 2$, it grows as $N^{2-\gamma}$. The full-energy argument above therefore does not give this rate in that range. This is a limitation of the proof, not a lower bound on learning. The larger entropy-compression proof in the earlier robust-prior paper covers those scales, with a much worse explicit polynomial. Obtaining quadratic teacher-width dependence under the same support condition remains open.

A A robust target cylinder at every width

We give the probability bound used in the statistical proof. It uses a robust digital implementation, not a claim that every implementation is stable. Let $N \geq C$. Partition the neurons into C balanced clone groups of sizes $k_a \in \{\lfloor N/C \rfloor, \lceil N/C \rceil\}$, so $C/2 \leq N/k_a \leq 2C$. Let P project onto vectors constant on each clone group. Fill unused logical slots with available constants or copies of an existing Boolean wire. The ideal wire vector $b^s(x)$ then has entries in $\{-1, 1\}$, belongs to the range of P, and has Euclidean norm $\sqrt{N}$.

Use gain A in the tanh simulation of every Boolean gate. For example, NAND on $u, v \in \{-1, 1\}$ is the sign of $1 - u - v$; copies and negations use u and $-u$. Bias constants in internal layers are supplied by counted constant wires. The ideal signed preactivation of every gate is at least A. In the clone implementation, each coefficient from a logical wire is divided equally among that wire's clones. Set

$$e_* = \frac{1}{16\sqrt{2C}}, \qquad \tau_0 = \frac{e_*}{4}, \qquad b_L = \sqrt{16 \log 9 + 8 \log(4L)},$$
$$A = \max\{32 b_L, 2 \log(8/e_*)\}, \quad a = \max\{2, 256c\}, \quad \rho = \min\{1/4, A/[4(d+1)], A e_*/(64\sqrt{C})\}.$$

The teacher readout averages the clones of the output wire with total coefficient a.

In Gaussian coordinates $\sqrt{d}\, W^1$, $\sqrt{N}\, W^s$, and Nv, all entries have variance t^2. Restrict each first-layer coordinate, each internal row's projection onto every normalized clone-group direction, and each readout coordinate to an interval of radius ρt about its teacher value. The internal matrices then decompose as

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \leq \rho t \sqrt{C}, \quad E^s P = 0.$$

The remaining Gaussian bulk E^s is independent of the constrained coordinates. A 1/4-net of the unit sphere has at most 9^N points, and the usual two-net Gaussian bound gives

$$\Pr\{\|E^s\|_{\mathrm{op}} > b_L t\} \leq 2 \exp(2N \log 9 - N b_L^2/8) \leq \frac{1}{2L}.$$

Thus all bulk norms are at most $b_L t$ with conditional probability at least 1/2.

Finite disturbance robustness. Allow a separate RMS disturbance of size τ_0 at each hidden layer and scalar output disturbance at most τ_0. This includes the entire total-budget ball $\|\eta\|_* \le \tau_0$. Write $e_s = \|h_\eta^s - b^s\|_2/\sqrt{N}$. If $e_{s-1} \le e_*$, every clone-group mean differs from its ideal bit by at most $e_*\sqrt{N/k_a} \le 1/16$. At most three teacher coefficients, including a constant coefficient, enter a gate, so its intended signed preactivation remains at least $A/2$. For $\epsilon \in \{-1, 1\}$ and real u, the elementary global secant bound

$$\epsilon q \ge A/2 \quad \Longrightarrow \quad |\tanh(q+u) - \epsilon| \le 2e^{-A/2} + 8|u|/A$$

holds: for $|u| < A/4$ use saturation, and otherwise the linear term is at least two. Since $E^s b^{s-1} = 0$ and $\|h_\eta^{s-1}\|_2/\sqrt{N} \le 1 + \tau_0$, it follows that

$$e_s \le 2e^{-A/2} + (8/A)\rho\sqrt{C}(1+\tau_0) + (8b_L/A)e_{s-1} + \tau_0 \le e_*.$$

Each of the four terms is at most $e_*/4$. The first layer initializes the same invariant: its coordinatewise preactivation perturbation is at most $\rho(d+1) \le A/4$, and its postactivation disturbance contributes at most τ_0 in RMS norm. Consequently the noisy signed readout is at least

$$f(x)z_w(x;\eta) \ge \frac{15a}{16} - \rho(1+\tau_0) - \tau_0 > 1 \qquad (x \in \mathcal{X},\ \|\eta\|_* \le \tau_0).$$

The readout center satisfies $\sqrt{N}\|v_*\| \ge a\sqrt{C/2}$, and its permitted perturbation has $\sqrt{N}\|v - v_*\| \le \rho$. Therefore $E_v \ge a^2 C/8$, whence

$$r_c(w) \le \frac{2\sqrt{2}c}{a\sqrt{C}} \le \frac{1}{64\sqrt{2C}} = \tau_0.$$

All clean and noisy clipped scores on the calibrated ball equal $f(x)/2$. The entire cylinder, with the bulk-norm event imposed, therefore belongs to $\mathcal{R}_{N,c}$ and satisfies the training likelihood for every sample. This also proves why every fixed $c > 0$ is allowed: a larger calibration requires only a polynomially larger readout coefficient.

Gaussian mass. There are Nq constrained independent scalar coordinates, where $q = d + 2 + (L-2)C$. The sum of their squared centers is at most NB_0, with

$$B_0 = 9dA^2 + 18(L-2)CA^2 + 2a^2C.$$

These loose constants cover the fan-in-two gate implementation above. For $Z \sim \mathcal{N}(0, t^2)$, $\rho \le 1/4$, and any center m,

$$\Pr\{|Z - m| \le \rho t\} \ge \frac{2\rho}{e\sqrt{2\pi}} \exp(-m^2/t^2).$$

Multiplication over the coordinates and the independent bulk event gives

$$\mu_{N,t}(\text{target cylinder}) \ge \exp(-ND_*/t^2), \qquad D_* = B_0 + q\log\frac{e\sqrt{2\pi}}{2\rho} + \log 2. \tag{3}$$

Here $D_* \ge 1$ and is bounded by a universal polynomial in d, L, C, c. The construction and estimate hold at every integer width $N \ge C$.

References

[1] ChatGPT-6 Astra and Claude (Fable 5 and Opus 5). *Depth dependence, internal robustness, and Bayesian neural-network learning*. Research report, 17 October 2026, Section 3, Theorem 3.1. Source and build instructions.

[2] ChatGPT-6 Astra, Claude Fable 5 and Opus 5, Kaarel Hänni, and Dmitry Vaintrob. *A polynomial-depth learning bound for NNBayes on robust networks*. Research manuscript, 18 October 2026. Source and build instructions.

[3] Pierre Tarrès and Yuan Yao. *Online learning as stochastic approximation of regularization paths: optimality and almost-sure convergence*. Appendix A, Corollary A.2, equation (A-1). https://arxiv.org/abs/1103.5538.